

Crossing-Point Analysis for Weighted Rolling Validation: Theory

Aileen Tao

1 Question and approach

Rolling validation (RV) selects between models by adding their validation losses over successive training sizes. The target, however, is the model with smaller population prediction risk at the *current* training size. If that preferred model changes as the training sample grows, the two comparisons need not change sign together: RV still includes losses from earlier, smaller training samples. The question of this report is whether weighting later validation observations can bring the expected RV crossing close to the population-risk crossing.

The two tasks are distinct. First, expected RV should change preference at approximately the right training size (*alignment*). Second, the realized, random RV comparison should be sufficiently concentrated around its mean (*concentration*). Weighting can improve the first task while making the second harder.

Why Analysis 08 is not enough

Analysis 08 considered a sample mean against an oracle that was always better in population risk. It studied the mean and variance of the RV loss difference and whether their ratio gave a selection-consistency bound. There was no preference reversal, hence no crossing to align. The present model deliberately has a reversal. Its nonzero mean parameter also creates a new linear random term in RV, so its variance cannot be imported from 08 without calculation.

Analysis 09 therefore introduces a crossing model in which the population-preferred model changes with training size. The report first studies the population crossing and the later ordinary-RV crossing. It then asks whether weighting can align the expected RV crossing while retaining enough concentration for reliable selection.

2 The two comparisons studied in 09

For models 0 and 1 trained on n observations, write

$$\Delta_R(n) = R_0(n) - R_1(n).$$

The population-risk difference $\Delta_R(n)$ compares the two models only at the *current* training size n . It is not accumulated over previous training sizes. By contrast, ordinary RV accumulates validation-loss differences from earlier training sizes. Its expected difference $\Delta_{RV}(n)$ therefore contains historical contributions even after current population risk has changed its preferred model.

A negative value favors Model 0, a positive value favors Model 1, and zero means equal population risks. If $RV_{k,w}(n)$ denotes Model k 's weighted RV loss through the validation time with n training observations, write

$$\Delta_w(n) = \mathbb{E}[RV_{0,w}(n) - RV_{1,w}(n)].$$

Its sign describes the preference of *expected* weighted RV, not necessarily a particular realized selection. Thus $\Delta_R(n)$ tells us what the population comparison prefers, while $\Delta_w(n)$ tells us what expected weighted RV prefers. We compare their first strict negative indices:

$$n_{\text{pop}} = \min\{n \geq 1 : \Delta_R(n) < 0\}, \quad n_{\text{RV},w} = \min\{n \geq 1 : \Delta_w(n) < 0\}.$$

The sets are shown to be nonempty below for the weights used here. The goal is to make these indices close, principally in the ratio sense as the crossing moves outward.

3 A model with a genuine crossing

Let

$$Y_i = \theta + \xi_i, \quad \xi_i \stackrel{\text{iid}}{\sim} N(0, 1), \quad 0 < |\theta| < 1.$$

Model 0 predicts a new response using the training sample mean $\bar{Y}_n$; Model 1 always predicts zero. Both use squared loss. Model 0 is unbiased but has estimation variance. Model 1 has no estimation variance but retains squared bias θ^2 . Hence the zero predictor can be better early, and the sample mean better later. A fixed shrinkage predictor would give a similar bias–variance crossing, but the zero predictor keeps the mechanism and formulas transparent.

3.1 Population prediction risk

Here $R_k(n)$ averages squared prediction loss over the training data and an independent new response. The exact crossing is as follows.

Theorem 3.1 (Population crossing).

$$R_0(n) = 1 + \frac{1}{n}, \quad R_1(n) = 1 + \theta^2, \quad \Delta_R(n) = \frac{1}{n} - \theta^2, \quad n_{\text{pop}} = \lfloor \theta^{-2} \rfloor + 1.$$

Interpretation. The common 1 is new-response noise. The comparison is between the sample mean’s estimation variance and the zero predictor’s squared bias. The displayed index is the first *strict* Model-0 preference; an equality at $n = \theta^{-2}$ is a tie.

Proof. See Appendix A.1.

3.2 Why ordinary RV crosses late

At validation time $m + 1$, Model 0 has been trained on m observations. Its expected loss difference relative to Model 1 is $1/m - \theta^2$. Ordinary RV accumulates these one-time differences:

$$\Delta_{\text{RV}}(n) = \sum_{m=1}^n \left(\frac{1}{m} - \theta^2 \right). \tag{1}$$

Define

$$H_n = \sum_{m=1}^n \frac{1}{m} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}.$$

H_n is the n -th harmonic number, and $\Delta_{\text{RV}}(n) = H_n - n\theta^2$. Since $H_n \sim \log n$, positive contributions from early validation times accumulate approximately logarithmically. The one-time expectation and sum are derived in Appendix A.2.

The distinction is temporal: $\Delta_R(n)$ compares population risks only at the current training size n , whereas $\Delta_{\text{RV}}(n)$ is an accumulated expected comparison. After the current population risk begins to favor Model 0, ordinary RV can still favor Model 1 because its sum retains positive contributions from earlier, smaller training sizes.

Proposition 3.2 (Ordinary-RV lag). *As $\theta \rightarrow 0$,*

$$n_{\text{RV}} \sim \theta^{-2} \log(\theta^{-2}), \quad \frac{n_{\text{RV}}}{n_{\text{pop}}} \sim \log(\theta^{-2}) \rightarrow \infty.$$

Interpretation. The accumulated history delays the expected ordinary-RV crossing by a growing factor relative to the population crossing.

Proof. See Appendix A.3.

4 Weighted RV and crossing alignment

The lag originates in accumulated early contributions. Increasing weights give more influence to later validations, which use larger training samples. For deterministic $w_i > 0$, independent of the data, define the realized weighted-RV loss difference and its expectation by

$$\Xi_{n,w} = \sum_{i=2}^{n+1} w_i \{(Y_i - \bar{Y}_{i-1})^2 - Y_i^2\}, \quad \Delta_w(n) = \mathbb{E}[\Xi_{n,w}].$$

A negative value selects Model 0. The final validation prediction uses n training observations. Multiplying all weights by one positive constant changes neither sign nor crossing.

At validation time i , the expected loss difference is

$$d_i = \frac{1}{i-1} - \theta^2.$$

Define the weighted historical reciprocal sum and the total weight:

$$A_n = \sum_{i=2}^{n+1} \frac{w_i}{i-1}, \quad W_n = \sum_{i=2}^{n+1} w_i.$$

The expected weighted-RV difference is

$$\Delta_w(n) = A_n - \theta^2 W_n. \tag{2}$$

Here A_n collects the estimation-variance terms from all validation times, while $\theta^2 W_n$ collects the bias terms. The derivation of d_i and (2) is in Appendix B.1.

Normalizing by W_n makes A_n/W_n the weighted average of the historical reciprocal training sizes. Population risk uses only the current value $1/n$. Their exact relationship is

$$\frac{\Delta_w(n)}{W_n} = \Delta_R(n) + \left(\frac{A_n}{W_n} - \frac{1}{n} \right). \tag{3}$$

The correction in parentheses is the effect of earlier training sizes. It is nonnegative, and strictly positive for $n > 1$ with positive weights, so the expected weighted-RV crossing cannot precede the population crossing. The identity and this crossing order are proved in Appendix B.2; crossing monotonicity and existence are established in Appendix B.3.

The alignment question is visible by comparing

$$\frac{\Delta_w(n)}{W_n} = \frac{A_n}{W_n} - \theta^2, \quad \Delta_R(n) = \frac{1}{n} - \theta^2.$$

Their only difference is A_n/W_n versus $1/n$. To make the expected weighted comparison resemble the current population comparison near their moving crossing, the target is $A_n/W_n \sim 1/n$, equivalently $nA_n/W_n \rightarrow 1$. The next theorem gives this condition a precise crossing interpretation.

Theorem 4.1 (Sufficient crossing alignment). *If the positive deterministic weights do not depend on θ and*

$$\frac{nA_n}{W_n} \rightarrow 1, \tag{4}$$

then $n_{\text{RV},w}$ exists for all sufficiently small nonzero θ and

$$\frac{n_{\text{RV},w}}{n_{\text{pop}}} \rightarrow 1 \quad (\theta \rightarrow 0).$$

Interpretation. If the weighted average of reciprocal training sizes approaches the latest reciprocal training size, the *expected* weighted-RV crossing aligns with the population crossing. The theorem does not control random fluctuations.

Proof. See Appendix B.4.

4.1 Representative weights

Crossing alignment requires $A_n/W_n \sim 1/n$, equivalently $nA_n/W_n \rightarrow 1$. The following families show how different rates of weight growth affect this condition.

Log-power weights. Consider $w_i = [\log(i+1)]^p$ for $p > 0$. These weights test whether a gradual emphasis on later observations can remove the lag. Appendix B.5 establishes

$$\frac{A_n}{W_n} \sim \frac{\log n}{(p+1)n}, \quad \frac{n_{\text{RV},w}}{n_{\text{pop}}} \sim \frac{\log(\theta^{-2})}{p+1}.$$

The target is $1/n$, but the additional factor $\log n/(p+1)$ diverges. These weights therefore fail alignment, and their crossing ratio diverges rather than tending to 1. The case $p = 0$ is ordinary unweighted RV, $p = 1$ is the log weight, and $p = 2$ is the squared-log weight. Increasing p reduces the finite crossing lag without removing its asymptotic divergence. The full asymptotic derivation is in Appendix B.5.

Log-log weight. The slower-growing choice $w_i = \log \log(i+e)$ checks whether a gentler increase behaves differently. It satisfies

$$\frac{A_n}{W_n} \sim \frac{\log n}{n}.$$

This still exceeds the $1/n$ target by a factor of $\log n$. Although it grows much more slowly than log-power weights, it also fails asymptotic alignment: its crossing has the same leading lag ratio as ordinary RV. The derivation is in Appendix B.6.

Geometric weights. Consider $w_i = q^i$ for $q > 1$. These weights grow much faster, so most of the total weight falls on the latest validation observations. Their training sizes are close to n , making their reciprocal training sizes close to $1/n$. Appendix B.7 proves

$$\frac{nA_n}{W_n} \rightarrow 1.$$

Thus geometric weights satisfy Theorem 4.1, and $n_{\text{RV},w}/n_{\text{pop}} \rightarrow 1$: the *expected* weighted-RV crossing aligns asymptotically with the population crossing. The same appendix calculates

$$\frac{\sum_{i=2}^{n+1} w_i^2}{W_n^2} \rightarrow \frac{q-1}{q+1} > 0.$$

The geometric weights are highly concentrated on a small number of late validation times, so they achieve crossing alignment but retain relatively little averaging across validation observations.

The representative weights reveal a contrast. Slowly increasing log-type weights spread weight across many validation times and reduce finite lags, but their historical average A_n/W_n remains too large relative to $1/n$ for asymptotic alignment. Geometric weights align the crossing by concentrating on the latest times, which reduces averaging. This motivates the next section.

5 Randomness near the moving crossing

The population crossing satisfies $n_{\text{pop}} \sim \theta^{-2}$ as $\theta \rightarrow 0$, so n/n_{pop} is approximately $n\theta^2$. The quantity $n\theta^2$ tells us where the current training size lies relative to the moving population crossing. Letting $\theta \rightarrow 0$ sends that crossing to larger sample sizes, while requiring

$$n\theta^2 \rightarrow c \in (0, \infty)$$

keeps n at a fixed relative position. Here $c < 1$ is before the population crossing, $c = 1$ is at its leading-order location, and $c > 1$ is after it. This is the relevant regime because the research question asks whether weighted RV behaves correctly near the point where the population-preferred model changes. The precise sign calculation is in Appendix C.1.

Crossing alignment concerns the expectation of weighted RV. The realized statistic $\Xi_{n,w}$ is random, so its variance also matters near the crossing. A sufficient Chebyshev concentration condition, when its mean is nonzero, is

$$\frac{\text{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \rightarrow 0.$$

This would control the probability of a sign error relative to the expected statistic.

The 08 Gaussian toy model used $\theta = 0$. Here $Y_i = \theta + \xi_i$ with $\theta \neq 0$. Expanding a validation loss difference produces an additional linear Gaussian term $-2\theta\xi_i$. It vanishes when $\theta = 0$, which is why it was absent in 08. After weighting and summing, its random contribution is $-2\theta \sum_{i=2}^{n+1} w_i \xi_i$, with variance $4\theta^2 \sum_{i=2}^{n+1} w_i^2$. Concentrating weights on the latest times can improve alignment, but also makes this noise large relative to the squared total weight.

The exact variance identity is

$$\text{Var}(\Xi_{n,w}) = 2 \sum_{t=1}^n a_t^2 + 4 \sum_{l=2}^{n+1} (l-1)b_l^2 + 4\theta^2 \sum_{i=2}^{n+1} w_i^2, \quad (5)$$

where $a_t = \sum_{m=t}^n w_{m+1}/m^2$ for $1 \leq t \leq n$ and $b_l = \sum_{m=l}^n w_{m+1}/m^2 - w_l/(l-1)$ for $2 \leq l \leq n+1$, with empty sums set to zero. The first two terms describe quadratic Gaussian noise; the last is the new linear noise. The loss expansion and variance proof are entirely in Appendices D.1 and D.2.

Crossing alignment requires increasing emphasis on the latest validation times, since only their reciprocal training sizes are close to the current population term $1/n$. Concentrating weight near the end reduces averaging across validation observations. The expected crossing may therefore align while the realized statistic remains too noisy near it. To describe the amount of averaging, consider

$$\frac{\sum_{i=2}^{n+1} w_i^2}{W_n^2}.$$

This measures concentration of the normalized weights: it is small when weight is spread across many observations and larger when only a few observations dominate. Its reciprocal is an effective number of observations. Thus divergence of $n \sum_i w_i^2 / W_n^2$ means the effective amount of averaging is much smaller than order n .

Theorem 5.1 (Near-crossing obstruction). *Under (4),*

$$n \frac{\sum_{i=2}^{n+1} w_i^2}{W_n^2} \longrightarrow \infty.$$

For every sequence $\theta \rightarrow 0$ and $n\theta^2 \rightarrow c \in (0, \infty) \setminus \{1\}$,

$$\frac{\text{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \longrightarrow \infty. \quad (6)$$

Interpretation. Near the moving crossing, alignment and Chebyshev concentration cannot hold simultaneously in this O9 model. Strong late weighting reduces averaging and makes the new linear Gaussian noise too large relative to the squared mean. The Chebyshev ratio therefore diverges rather than vanishes. This criterion does not determine the limiting wrong-selection probability. The complete proof and variance calculation are in Appendices C.2–C.3 and D.1–D.2.

Logarithmic and log-log weights. These spread mass across more validation times and retain better averaging, but their expected crossings remain asymptotically late. For $c > 1$, they can concentrate around an expected sign that disagrees with current population preference.

Geometric weights. These align the expected crossing by concentrating strongly on the latest validation times, but the same concentration causes the near-crossing Chebyshev ratio to diverge.

For fixed $\theta \neq 0$ and $n \rightarrow \infty$, some intermediate-growth weights can achieve both alignment and concentration, but this is a different and weaker regime than the moving-crossing problem.

6 What O9 establishes

Analysis O9 shows that ordinary rolling validation changes model preference later than population risk because it accumulates information from earlier training sizes. Weighting can reduce this lag, and sufficiently concentrated late-stage weights can align the expected crossing with the population crossing. Near the moving crossing, however, such concentration reduces averaging and prevents the Chebyshev concentration condition from holding. In this model, the same positive deterministic weighting behavior cannot achieve both properties.

7 What remains open

The result is specific to the 09 model, positive deterministic weights, and the near-crossing Chebyshev argument. It does not determine the exact limiting wrong-selection probability. Future work could study that probability directly or examine other weighting schemes, validation designs, or crossing models.

A Population risk and ordinary RV

A.1 Population-risk formulas and strict crossing

Let $Y_{\text{new}} = \theta + \xi_{\text{new}}$ be independent of the training sample. Since $\bar{Y}_n = \theta + S_n/n$ with $S_n = \sum_{t=1}^n \xi_t$, the Model-0 error is $\xi_{\text{new}} - S_n/n$. Independence, zero means, and $\text{Var}(S_n) = n$ give $R_0(n) = 1 + 1/n$. Model 1 has error $\theta + \xi_{\text{new}}$, hence $R_1(n) = 1 + \theta^2$. Their difference is $1/n - \theta^2$, negative exactly when $n > \theta^{-2}$. The least integer strictly above θ^{-2} is $\lfloor \theta^{-2} \rfloor + 1$, including when θ^{-2} itself is an integer (the equality index is a tie). This proves Theorem 3.1.

A.2 One-time and ordinary-RV expectation

At time $m + 1$, $\bar{Y}_m = \theta + S_m/m$ and the fresh error of Model 0 is $\xi_{m+1} - S_m/m$; that of Model 1 is $\theta + \xi_{m+1}$. Expanding and using independence of ξ_{m+1} and S_m gives

$$\mathbb{E}[(Y_{m+1} - \bar{Y}_m)^2 - Y_{m+1}^2] = (1 + 1/m) - (1 + \theta^2) = 1/m - \theta^2.$$

Summing over $m = 1, \dots, n$ proves (1).

A.3 Ordinary-RV crossing asymptotics

Put $h = \theta^2$ and $N = h^{-1}$. The mean $H_n - nh$ has increments $1/(n+1) - h$, so it is initially increasing and eventually decreasing to $-\infty$; its first strict negative index exists and is unique. Integral bounds give $\log(n+1) \leq H_n \leq 1 + \log n$. Fix $\varepsilon \in (0, 1)$ and set

$$n_- = \lfloor (1 - \varepsilon)N \log N \rfloor, \quad n_+ = \lceil (1 + \varepsilon)N \log N \rceil.$$

Both tend to infinity; $\log n_{\pm} = \log N + \log \log N + O(1)$, while $n_{\pm}h = (1 \pm \varepsilon) \log N + o(1)$. Thus $H_{n_-} - n_-h = \varepsilon \log N + o(\log N) > 0$ and $H_{n_+} - n_+h = -\varepsilon \log N + o(\log N) < 0$. The crossing lies between n_- and n_+ . Divide by $N \log N$ and let $\varepsilon \downarrow 0$ to obtain $n_{\text{RV}} \sim N \log N$. Since $n_{\text{pop}} = N + O(1)$, Proposition 3.2 follows.

B Weighted RV and crossing alignment

B.1 Weighted expectation

Apply the one-time expectation in Appendix A.2 with $m = i - 1$. Deterministic weights and linearity of expectation give

$$\Delta_w(n) = \sum_{i=2}^{n+1} w_i ((i-1)^{-1} - \theta^2) = \sum_{i=2}^{n+1} \frac{w_i}{i-1} - \theta^2 \sum_{i=2}^{n+1} w_i = A_n - \theta^2 W_n.$$

No independence between validation losses is asserted or needed.

B.2 Exact remainder and crossing order

Divide the preceding identity by $W_n > 0$ and add and subtract $1/n$ to obtain (3). Each $1/(i-1) \geq 1/n$. Their positive-weighted average therefore satisfies $A_n/W_n \geq 1/n$; for $n > 1$, at least one earlier term is strictly larger and has positive weight, so the inequality is strict. If $\Delta_R(n) \geq 0$, then $\Delta_w(n)/W_n \geq 0$. Consequently a strict weighted-RV crossing cannot precede n_{pop} . Equality is possible at $n = 1$ or if nonnegative (rather than positive) weights are supported only on the final term.

B.3 Monotonicity and crossing existence

Write $r_n = A_n/W_n$. Adding the next validation time gives

$$r_{n+1} = \frac{W_n r_n + w_{n+2}/(n+1)}{W_n + w_{n+2}}.$$

Every term in r_n is at least $1/n > 1/(n+1)$, so $r_{n+1} < r_n$ for positive weights. Thus the equation $r_n < h$ has at most one first strict crossing. It exists exactly when $\inf_n r_n < h$; since r_n decreases, this is equivalent to $\lim_n r_n < h$. If the limit equals h , strict crossing need not occur. Under (4), $r_n \sim 1/n \rightarrow 0$, so crossing exists for every fixed $h > 0$; $r_1 = 1 > h$ because $0 < h < 1$.

B.4 Sufficient-alignment theorem

Let $h = \theta^2 \downarrow 0$ and fix $\varepsilon \in (0, 1)$. Set $n_- = \lfloor (1 - \varepsilon)/h \rfloor$ and $n_+ = \lceil (1 + \varepsilon)/h \rceil$. Both grow, and (4) gives $r_{n_{\pm}} = (1 + o(1))/n_{\pm}$. Therefore

$$\frac{r_{n_-}}{h} \rightarrow \frac{1}{1 - \varepsilon} > 1, \quad \frac{r_{n_+}}{h} \rightarrow \frac{1}{1 + \varepsilon} < 1.$$

By Appendix B.3, the first strict crossing lies in $(n_-, n_+]$ for small h . Rounding changes either endpoint by at most one. Divide by h^{-1} and let $\varepsilon \downarrow 0$; since $n_{\text{pop}} = h^{-1} + O(1)$, the crossing ratio tends to one. This proves Theorem 4.1.

B.5 Log-power weights and their crossings

Let $w_i = [\log(i+1)]^p$ with fixed $p \geq 0$. A shift by one changes $\log(i+1)$ by $O(1/i)$, so its p th power has the same leading order as $(\log i)^p$ (with $p = 0$ interpreted as 1). Integral comparison, or summation by parts, yields

$$A_n \sim \int_2^n \frac{(\log x)^p}{x} dx = \frac{(\log n)^{p+1}}{p+1}.$$

For completeness, the summand is eventually decreasing: its logarithmic derivative is $(p/\log x - 1)/x < 0$ for $x > e^p$. Thus its sum differs from the integral by a bounded initial contribution and an error no larger than a fixed eventual first term. The shift error is summable for $p \geq 0$. Also, for any $\delta \in (0, 1)$, the last $(1 - \delta)n$ indices have weights $(\log n)^p(1 + o(1))$ uniformly, while the first δn contribute at most $\delta n(\log n + O(1))^p$. Taking $n \rightarrow \infty$ and then $\delta \downarrow 0$ gives $W_n \sim n(\log n)^p$. Hence

$$r_n \sim \frac{\log n}{(p+1)n}.$$

For the crossing let $N = h^{-1}$ and $g_N = N \log N/(p+1)$. At $n_{\pm} = \lfloor (1 \mp \varepsilon)g_N \rfloor$ and $\lceil (1 \pm \varepsilon)g_N \rceil$, respectively, $\log n_{\pm}/\log N \rightarrow 1$, so $r_{n_{\pm}}/h \rightarrow (1 \mp \varepsilon)^{-1}$. Monotonicity from Appendix B.3 brackets the first crossing between them. Letting $\varepsilon \downarrow 0$ gives $n_{\text{RV},w} \sim g_N$ and, since $n_{\text{pop}} \sim N$, the ratio stated in the main text. This includes $p = 0$.

B.6 Log-log weight

Set $f(x) = \log \log(x + e) > 0$. Its derivative is $f'(x) = 1/[(x + e) \log(x + e)]$. For large x , $f(x) \sim \log \log x$ and $f(x)/x$ is decreasing, so integral comparison applies. The shift from $f(x)$ to $\log \log x$ produces an integrable error after division by x : the mean-value theorem bounds it by $C/(x^2 \log x)$ for sufficiently large x . Substituting $u = \log x$,

$$A_n = \int^n \frac{\log \log x}{x} dx + O(1) = \log n \log \log n - \log n + O(1) \sim \log n \log \log n.$$

The same last-fraction argument as in Appendix B.5, using $f(\delta n)/f(n) \rightarrow 1$ for fixed $\delta > 0$, gives $W_n \sim n \log \log n$. Hence $r_n \sim (\log n)/n$. Bracketing at $n_{\pm} = (1 \pm \varepsilon)N \log N$ exactly as in Appendix B.5 gives $n_{\text{RV},w} \sim N \log N$ and $n_{\text{RV},w}/n_{\text{pop}} \sim \log N$.

B.7 Geometric-weight alignment and effective averaging

For $w_i = q^i$ with $q > 1$, write $i = n + 1 - j$. Division by W_n makes the lag j have probabilities proportional to q^{-j} , $j = 0, \dots, n - 1$. The denominator $\sum_{j=0}^{n-1} q^{-j} \rightarrow q/(q - 1)$ and the numerator for nr_n is

$$\sum_{j=0}^{n-1} q^{-j} \frac{n}{n - j}.$$

For $j \leq n/2$, $n/(n - j) \leq 2$ and tends to 1 for each fixed j ; dominated convergence for the summable geometric series gives the same limit $q/(q - 1)$. For $j > n/2$, the sum is at most $n \sum_{j > n/2} q^{-j} \rightarrow 0$. Thus $nr_n \rightarrow 1$. Direct finite geometric sums also give

$$\frac{\sum_{i=2}^{n+1} q^{2i}}{(\sum_{i=2}^{n+1} q^i)^2} \rightarrow \frac{q - 1}{q + 1}.$$

Theorem 4.1 supplies the crossing ratio; the displayed square-weight limit measures its bounded effective averaging.

C Near-crossing alignment–concentration obstruction

C.1 The near-crossing scaling

Put $h = \theta^2$. Theorem 3.1 gives $n_{\text{pop}} = h^{-1} + O(1)$. Thus $nh \rightarrow c$ places n at asymptotic fraction c of the population crossing. Moreover $n\Delta_R(n) = 1 - nh \rightarrow 1 - c$, proving the before/after sign interpretation for $c \neq 1$.

C.2 Weight concentration and effective sample size

Let $p_{n,i} = w_i/W_n$, so $\sum_i p_{n,i} = 1$. Alignment is

$$\sum_{i=2}^{n+1} p_{n,i} \left(\frac{n}{i-1} - 1 \right) \rightarrow 0.$$

All summands are nonnegative. For fixed $\varepsilon \in (0, 1)$ and $i - 1 \leq (1 - \varepsilon)n$, the factor is at least $\varepsilon/(1 - \varepsilon)$. Hence the total mass there is at most $(1 - \varepsilon)/\varepsilon$ times the displayed sum, and tends to

zero. The remaining set has at most $\varepsilon n + 1$ indices and mass tending to one. Cauchy–Schwarz on that set gives

$$\sum_i p_{n,i}^2 \geq \frac{(\sum_{i-1 > (1-\varepsilon)n} p_{n,i})^2}{\varepsilon n + 1}, \quad \text{so} \quad \liminf_n n \sum_i p_{n,i}^2 \geq \frac{1}{\varepsilon}.$$

For any $K > 0$, choose $\varepsilon < 1/K$; the lower limit is then at least K . Thus $n \sum_i p_{n,i}^2 \rightarrow \infty$.

C.3 Proof of the obstruction theorem

Under alignment, $r_n = (1 + o(1))/n$. If $nh \rightarrow c \neq 1$, then by (2)

$$\frac{\mathbb{E}\Xi_{n,w}}{W_n} = r_n - h = \frac{1 - c + o(1)}{n}.$$

Appendix D.2 proves that the quadratic variance is nonnegative and the additional linear variance equals $4h \sum_i w_i^2$. Therefore

$$\frac{\text{Var}(\Xi_{n,w})}{(\mathbb{E}\Xi_{n,w})^2} \geq \frac{4h \sum_i p_{n,i}^2}{(r_n - h)^2} = \frac{4(c + o(1)) n \sum_i p_{n,i}^2}{(1 - c + o(1))^2} \rightarrow \infty$$

by Appendix C.2. The mean is positive when $c < 1$ and negative when $c > 1$, for sufficiently large n . The proof establishes failure of a sufficient Chebyshev argument, not a positive lower bound on wrong-selection probability. This proves Theorem 5.1.

C.4 The exact boundary

If $nh \rightarrow 1$ under alignment, $r_n - h = o(1/n)$. When this difference is zero, the variance-to-mean-squared ratio is undefined. When it is nonzero, the preceding lower bound has numerator $4h \sum p_{n,i}^2$ and denominator $o(n^{-2})$; writing $r_n - h = \delta_n/n$ with $\delta_n \rightarrow 0$ gives a lower bound $4(1 + o(1))n \sum p_{n,i}^2 / \delta_n^2 \rightarrow \infty$. No leading-order population sign is assigned at $c = 1$.

D Exact variance and a fixed-parameter comparison

D.1 Loss expansion and centering

Write $S_m = \sum_{t=1}^m \xi_t$. Since $\bar{Y}_m = \theta + S_m/m$,

$$(Y_{m+1} - \bar{Y}_m)^2 - Y_{m+1}^2 = (\xi_{m+1} - S_m/m)^2 - (\theta + \xi_{m+1})^2 = \frac{S_m^2}{m^2} - 2\xi_{m+1} \frac{S_m}{m} - 2\theta\xi_{m+1} - \theta^2.$$

The first quadratic term has mean $1/m$ and the other random terms have mean zero. Therefore, with

$$Q_{n,w} = \sum_{m=1}^n w_{m+1} \left\{ \frac{S_m^2}{m^2} - 2\xi_{m+1} \frac{S_m}{m} \right\},$$

we have $\Xi_{n,w} - \mathbb{E}\Xi_{n,w} = Q_{n,w} - 2\theta \sum_{i=2}^{n+1} w_i \xi_i$. This also independently verifies (2).

D.2 Gaussian orthogonality and coefficients

Expand $S_m^2 - m = \sum_{t=1}^m (\xi_t^2 - 1) + 2 \sum_{1 \leq t < u \leq m} \xi_t \xi_u$. Collecting each centered square gives coefficient

$$a_t = \sum_{m=t}^n \frac{w_{m+1}}{m^2}, \quad 1 \leq t \leq n.$$

For each pair $t < l$, the square expansion contributes $2 \sum_{m=l}^n w_{m+1}/m^2$; the validation term at $m = l - 1$ contributes $-2w_l/(l - 1)$. Thus, with empty sums zero,

$$b_l = \sum_{m=l}^n \frac{w_{m+1}}{m^2} - \frac{w_l}{l - 1}, \quad 2 \leq l \leq n + 1,$$

and

$$Q_{n,w} = \sum_{t=1}^n a_t (\xi_t^2 - 1) + 2 \sum_{l=2}^{n+1} b_l \sum_{t=1}^{l-1} \xi_t \xi_l.$$

Independent standard Gaussians satisfy $\mathbb{E}(\xi_t^2 - 1)^2 = 2$ and $\mathbb{E}(\xi_t \xi_l)^2 = 1$ for $t \neq l$. Distinct centered squares, distinct unordered pair products, and a centered square against a pair product have covariance zero: in every cross-product, some independent centered Gaussian appears to an odd power or as a mean-zero factor. Each quadratic monomial also has zero covariance with every ξ_i , because every centered third Gaussian moment is zero. Hence the quadratic and linear parts are orthogonal, and

$$\text{Var}(Q_{n,w}) = 2 \sum_{t=1}^n a_t^2 + 4 \sum_{l=2}^{n+1} (l - 1) b_l^2, \quad \text{Var}\left(-2\theta \sum_{i=2}^{n+1} w_i \xi_i\right) = 4\theta^2 \sum_{i=2}^{n+1} w_i^2.$$

Adding proves the exact identity (5).

D.3 Slow-weight variance orders

Here $h = \theta^2$ and $nh \rightarrow c \in (0, \infty)$; consider $w_{m+1} = f(m)$, where $f(m) = 1$, $[\log(m + 2)]^p$ for fixed $p > 0$, or $\log \log(m + 1 + e)$. In each case $f(n)$ is slowly varying, and splitting sums at δn as in Appendix B shows

$$\sum_{m=1}^n f(m)^2 \sim n f(n)^2.$$

We verify the quadratic order without borrowing it from 08. The coefficients a_t are bounded by $\sum_{m=t}^\infty f(m)/m^2$. For the listed f , integral comparison and eventual monotonicity of $f(x)/x^{3/4}$ give $\sum_{m=t}^\infty f(m)/m^2 \leq C f(t)/t$ for large t . Consequently $\sum_t a_t^2 \leq C + C \sum_t f(t)^2/t^2 = O(1)$.

Set $T_n = \sum_{m=n+1}^\infty f(m)/m^2$ and $B_l = \sum_{m=l}^\infty f(m)/m^2 - f(l - 1)/(l - 1)$, so $b_l = B_l - T_n$ for $2 \leq l \leq n$ and $b_{n+1} = -f(n)/n$ (the harmless one-step indexing changes the following estimates only by lower-order terms). Integral comparison, summation by parts, and the mean-value theorem give

$$T_n \sim f(n)/n, \quad |B_l| \leq C \left(\frac{\sup_{x \in [l-1, l+1]} |f'(x)|}{1} + \frac{f(l)}{l^2} \right)$$

for smooth extensions of the listed f . To see the derivative bound, subtract the constant- $f(l)$ sum, whose discrepancy from $f(l)/(l - 1)$ is $O(f(l)/l^2)$; summation by parts bounds the remaining variation by $C \int_l^\infty |f'(x)| x^{-1} dx$. This is $O((\log l)^{p-1}/l)$ for log powers and $O(1/(l \log l))$ for log-log;

the constant case has zero derivative. Therefore $D_n = \sum_{l=2}^n (l-1)B_l^2$ is $O(1)$ for $f = 1$ and log-log, while for log powers it is bounded by

$$C + C \sum_{l=3}^n \frac{(\log l)^{2p-2}}{l} = \begin{cases} O(1), & 0 < p < 1/2, \\ O(\log \log n), & p = 1/2, \\ O((\log n)^{2p-1}), & p > 1/2. \end{cases}$$

The three cases follow by $u = \log x$ in the comparison integral $\int^n (\log x)^{2p-2} dx/x$. In every case $D_n = o(f(n)^2)$ when $f(n) \rightarrow \infty$; for $f = 1$, $D_n = O(1)$ and the exact $B_l = O(l^{-2})$ is handled below. The terminal contribution is

$$4T_n^2 \sum_{l=2}^n (l-1) \sim 4 \frac{f(n)^2}{n^2} \frac{n^2}{2} = 2f(n)^2.$$

Expanding $(B_l - T_n)^2$, Cauchy-Schwarz bounds the magnitude of its cross term by $2\{D_n T_n^2 \sum_{l=2}^n (l-1)\}^{1/2} = o(f(n)^2)$ for the growing weights. The end coefficient contributes $4n(f(n)/n)^2 = o(f(n)^2)$. Thus $\text{Var}(Q_{n,w}) \sim 2f(n)^2$ for the growing log and log-log weights. For $f = 1$, direct integral comparison yields $B_l = O(l^{-2})$. Thus $D_n = O(1)$ and $\sum_{l=2}^n (l-1)|B_l| = O(\log n)$. Since $T_n = O(1/n)$, the cross term is $O(\log n/n) = o(1)$; the end coefficient is $O(1/n)$. Therefore $\text{Var}(Q_{n,w}) \sim 2$ also in the unweighted case.

The linear variance is $4h \sum_i w_i^2 \sim 4cf(n)^2$. Hence the full variance is of order $f(n)^2$ (indeed asymptotic to $(2 + 4c)f(n)^2$ for the growing weights). Meanwhile Appendix B gives $\mathbb{E}\Xi_{n,w} \sim (\log n)f(n)/(p+1)$ for log powers and $\sim (\log n)f(n)$ for log-log; the $-hW_n$ term is only $O(f(n))$. The ordinary-weight mean is $\sim \log n$ and its full variance tends to $2 + 4c$. Therefore all four Chebyshev ratios have order $(\log n)^{-2}$. Their means are positive for fixed c , establishing the sign statement in the main text.

D.4 Stretched-exponential weights at fixed θ

Let $w_i = \exp(i^\beta)$ with $0 < \beta < 1$. Put $F(x) = e^{x^\beta}$. Substitution $u = x^\beta$ and integration by parts yield, for $a = 1, 2$,

$$\int^n F(x)^a dx = \frac{1}{\beta} \int^{n^\beta} e^{au} u^{1/\beta-1} du \sim \frac{n^{1-\beta} e^{an^\beta}}{a\beta}.$$

Because F is increasing, bounding sums between adjacent integrals changes them by at most $O(F(n)^a)$, negligible relative to $n^{1-\beta}F(n)^a$. Endpoint shifts by one are negligible since $(n+1)^\beta - n^\beta \rightarrow 0$. Therefore

$$W_n \sim \frac{n^{1-\beta} e^{n^\beta}}{\beta}, \quad \sum_i w_i^2 \sim \frac{n^{1-\beta} e^{2n^\beta}}{2\beta}, \quad \frac{\sum_i w_i^2}{W_n^2} \sim \frac{\beta}{2n^{1-\beta}} \rightarrow 0.$$

These sums show an effective terminal window of order $n^{1-\beta} = o(n)$. For any fixed $\delta > 0$, the total weight at $i-1 \leq (1-\delta)n$ is at most $ne^{((1-\delta)n+O(1))^\beta} = o(W_n)$, because the exponential gap is of order n^β . On the remaining indices, $n/(i-1) \leq 1/(1-\delta)$; for the earlier indices their weighted contribution to nr_n is negligible by the same exponential gap even after the factor $n/(i-1) \leq n$. Hence $1 \leq \liminf nr_n \leq \limsup nr_n \leq 1/(1-\delta)$, and $\delta \downarrow 0$ proves alignment.

For fixed $\theta \neq 0$, $\mathbb{E}\Xi_{n,w}/W_n = r_n - \theta^2 \rightarrow -\theta^2$. To bound the quadratic variance, define the centered single-time quadratic increment $U_m = (S_m^2 - m)/m^2 - 2\xi_{m+1}S_m/m$. Gaussian moments

and independence give $\|U_m\|_2 \leq \sqrt{2}/m + 2/\sqrt{m} \leq C/\sqrt{m}$. Minkowski's inequality and then Cauchy–Schwarz yield

$$\left\| \frac{Q_{n,w}}{W_n} \right\|_2 \leq C \sum_{m=1}^n \frac{w_{m+1}}{W_n \sqrt{m}} \leq C \left(\sum_{m=1}^n \frac{w_{m+1}}{W_n m} \right)^{1/2} = C \sqrt{r_n} \rightarrow 0.$$

The normalized linear variance is $4\theta^2 \sum_i w_i^2 / W_n^2 \rightarrow 0$. Orthogonality from Appendix D.2 then gives $\text{Var}(\Xi_{n,w}) / W_n^2 \rightarrow 0$, and division by the mean-squared limit θ^4 proves fixed- θ concentration. This does not apply to $\theta \rightarrow 0$ with $n\theta^2 \rightarrow c$.